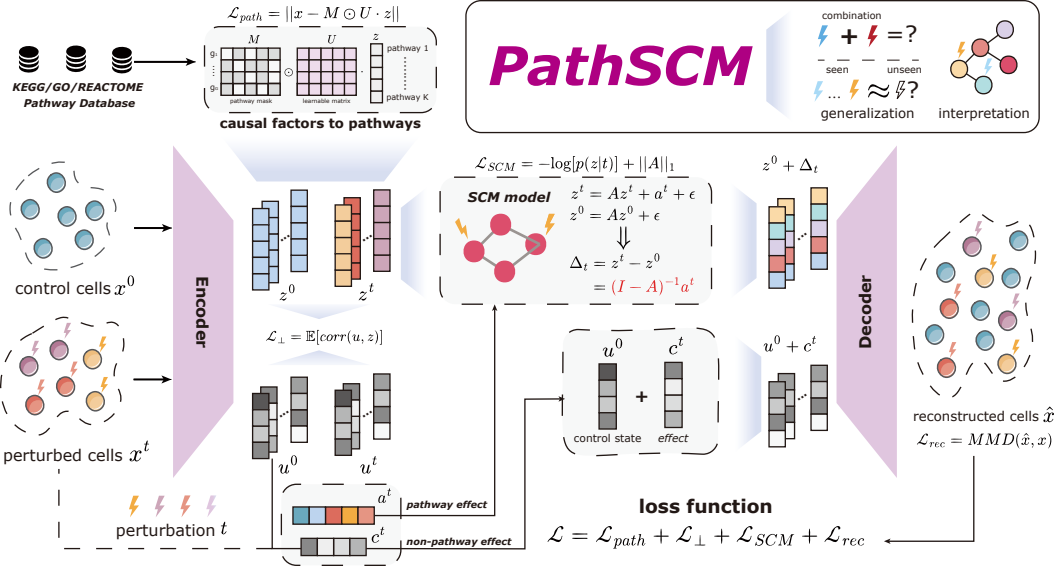

PathSCM
KEGG/GO/REACTOME
Pathway Database
$\mathcal{L}_{path} = ||x - M \odot U \cdot z||$
M
U
z
pathway 1
pathway K
pathway mask
learnable matrix
causal factors to pathways
combination
seen
unseen
generalization
interpretation
control cells x^0
perturbed cells x^t
perturbation t
Encoder
z^0
z^t
$\mathcal{L}_{SCM} = -\log[p(z|t)] + ||A||_1$
SCM model
$z^t = Az^t + a^t + \epsilon$
$z^0 = Az^0 + \epsilon$
$\Delta_t = z^t - z^0$
$= (I - A)^{-1} a^t$
$z^0 + \Delta_t$
$\mathcal{L}_{\perp} = \mathbb{E}[corr(u, z)]$
u^0
u^t
a^t
pathway effect
c^t
non-pathway effect
control state
effect
$u^0 + c^t$
Decoder
reconstructed cells $\hat{x}$
$\mathcal{L}_{rec} = MMD(\hat{x}, x)$
loss function
$\mathcal{L} = \mathcal{L}_{path} + \mathcal{L}_{\perp} + \mathcal{L}_{SCM} + \mathcal{L}_{rec}$